

**Scalable Demographic Bias Auditing in Data Preprocessing Pipeline
using a Hybrid WOA-DE Algorithm for Optimized Metadata Provenance**

A Thesis

Presented to the Faculty of the
College of Computer and Information Sciences
Polytechnic University of the Philippines

In Partial Fulfilment of the Requirements for the Degree
Bachelor of Science in Computer Science

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May 2026

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CHAPTER 1

THE PROBLEM AND ITS SETTING

This chapter contains the introduction, the theoretical and conceptual framework, statement of the problem, hypothesis, the significance of this study, scope and limitation, and operational definitions.

Introduction

Artificial Intelligence has become a cornerstone of automated decision-making in modern society, affecting critical areas such as banking, hiring, and healthcare (Mehrabi et al., 2021). As these systems become more integrated into daily life, a major trend in computing has shifted toward ensuring algorithmic fairness. However, an AI system is only as fair as the data it receives. Demographic bias, the overrepresentation or underrepresentation of specific groups based on protected attributes like race, sex, or age often results in data that fails to reflect the diverse needs of the broader population. When these biases persist, they contaminate business processes and lead to products that are inadvertently discriminatory. Real-world consequences range from facial recognition systems struggling to identify people of color to automated hiring tools that exclude qualified candidates, ultimately eroding public trust and undermining the efficacy of AI-driven decisions (Dominguez-Catena et al., 2025). If the underlying data contains unfair patterns or demographic biases, the AI will inevitably produce biased outcomes that have a negative real-world impact (Suresh and Gutttag, 2021).

Before an AI model begins the learning process, the data must pass through a preprocessing pipeline, a stage where raw data is cleaned, transformed, and engineered. This stage is critical because if the data is affected by bias at this level, the

final AI model will be fundamentally flawed (Pagano et al., 2023). A significant challenge in this field is that bias is often silent, meaning it hides deep within complex cleaning and transformation sequences where it cannot be easily detected. Currently, existing tools often lack the speed to handle large datasets or the level of detail required to locate these hidden biases (Pessach and Shmueli, 2022).

This tracking challenge introduces a highly complex computing problem: combinatorial optimization within massive, multi-dimensional search spaces. When data undergoes multiple preprocessing steps (e.g., missing value imputation, scaling, encoding), each step alters the distribution of protected attributes. Finding the exact combination of pipeline configurations that minimizes demographic bias while maintaining data utility results in an exponential explosion of potential solutions. To track these transformations, this study utilizes metadata provenance, which acts as a systematic record that logs every change and movement of data throughout its lifecycle (Ormenisan et al., 2020). While a provenance log is essential because it allows researchers to trace data back through its processing steps, searching through these massive, high-dimensional records to find the optimal configuration to audit bias is a computationally expensive task that standard exact algorithms cannot solve within a reasonable timeframe.

To navigate this intractable search space, metaheuristic optimization algorithms are required. Two prominent population-based algorithms are the Whale Optimization Algorithm (WOA) and Differential Evolution (DE). WOA, which simulates the bubble-net hunting behavior of humpback whales, possesses exceptional global exploration capabilities due to its random search and spiral update mechanisms, allowing it to quickly scan vast areas of the search space. However, WOA frequently suffers from

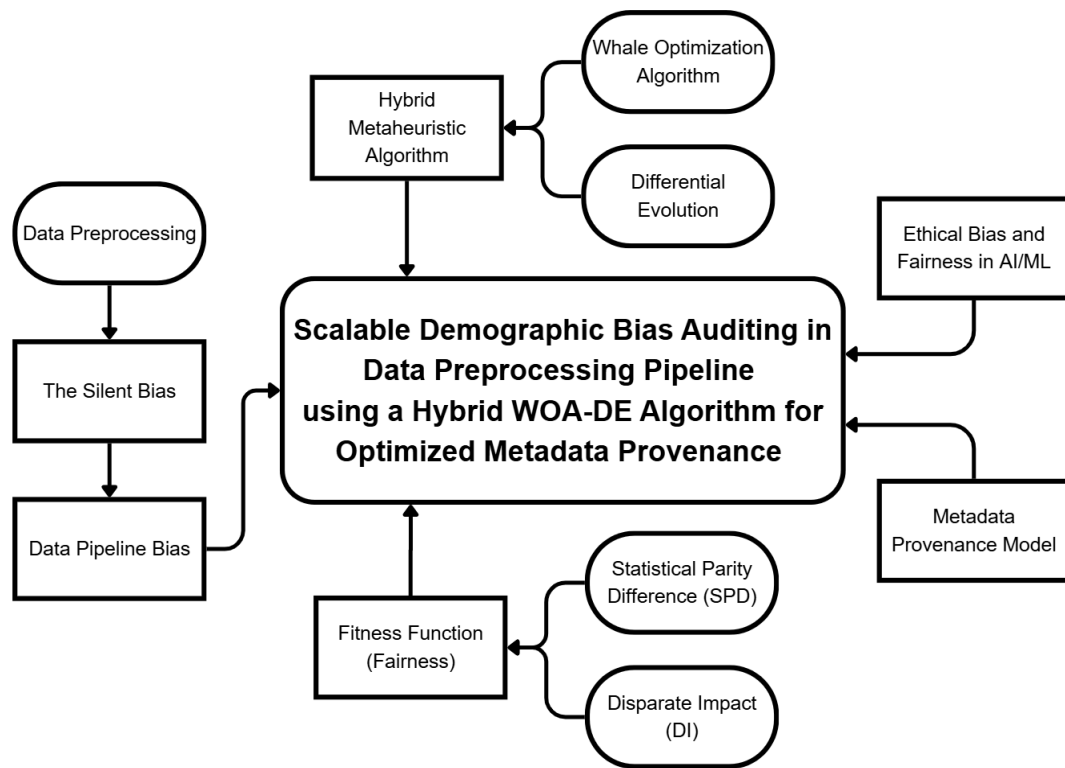
premature convergence and poor local exploitation, often getting trapped in local optima when refining specific pipeline parameters. Conversely, DE mimics biological evolution through mutation, crossover, and selection. It excels at local exploitation and maintaining population diversity, but it suffers from a slow convergence rate during the early stages of a global search. By creating a hybrid WOA-DE algorithm, their algorithmic behaviors complement each other perfectly: WOA rapidly narrows down the vast metadata provenance search space to locate promising regions, while DE steps in to prevent premature convergence, maintaining population diversity and precisely fine-tuning the parameters to discover the global optimum.

Importantly, this optimized pipeline focuses strictly on bias auditing and root-cause analysis rather than fully automated mitigation. While automatic mitigation sounds efficient, it remains highly discouraged in algorithmic fairness frameworks. Bias is not merely a statistical anomaly that can be coded away; it is deeply intertwined with social, cultural, and contextual nuances that an automated system cannot comprehend. Automated correction algorithms often introduce blind spots, inadvertently stripping away valid, non-biased variance or creating a false sense of equity that masks deeper systemic disparities. Furthermore, automated mitigation removes human oversight, stripping domain experts of accountability and the contextual understanding necessary to make ethical decisions. By prioritizing descriptive auditing over prescriptive automated correction, this system empowers human auditors with precise metadata insights to understand why and where bias occurs, ensuring meaningful, legally compliant, and sociotechnically sound interventions.

Consequently, this research seeks to fill the current gap in the industry by introducing Scalable Demographic Bias Auditing in Data Preprocessing Pipeline using a Hybrid WOA-DE Algorithm for Optimized Metadata Provenance.

Theoretical Framework

Figure 1: Theoretical Framework



Data Pipeline Bias and The Silent Bias

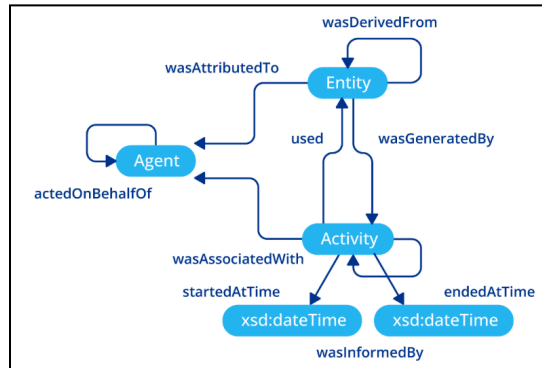
Kumar-B et al. (2021) investigated how bias enters and propagates through Natural Language Processing (NLP) and Machine Learning (ML) data pipelines. Their study reveals that the data pipeline itself is a primary source of

bias or unfairness. It identified multiple stages where bias can occur such as: data collection, in which sources may overrepresent or underrepresent groups. Data annotation, in which human annotators bring their own subjective opinions or biases. Data preprocessing or cleaning, in which certain groups are unintentionally excluded. Feature representation, in which embeddings can encode historical or social biases. Model training and evaluation, in which bias from previous pipeline propagates to prediction. Lastly, Applications, in which biased outcomes become visible to the users. This supports the researcher's claim that bias is a persistent issue throughout the data pipeline

An article by Trivedi, J. (2025) corroborates the "Data Pipeline Bias" as he highlighted that bias enters earlier in the data pipeline silently. This includes the early stages of: data collection, data annotation or labelling, data cleaning, and feature representation. The bias is considered "silent" because it is rarely recognized at the time it is introduced in the pipeline. Furthermore, when the silent bias propagates further downstream in the pipeline, it is harder to detect and correct later in the process. Both studies by Kumar-B and Trivedi confirm that biases do really exist from the beginning to the end of the data pipeline.

Metadata Provenance Model

Figure 2: Metadata Provenance Model



Metadata provenance works by making systems record data origin, history, and transformation. A diagram by Khan, U. H. (2025) demonstrates how data flows through transformation and how it is recorded. This tracks where the data came from, what processes are used to change it, the one responsible for the transformation, and when did it happen. This study is relevant as it proves how data provenance works and why it's useful and trustworthy in detecting biases. In this thesis, metadata provenance is used to record the different transformations a data goes through in the data preprocessing pipeline, such as removing duplicates, handling missing values, and handling outliers.

Ethical Bias & Fairness in AI/ML

Bias in AI systems come from multiple sources, such as biased training data, algorithm design choices, and human decision-making (Ferrara, 2024). It affects different domains such as healthcare, employment, and finance. Furthermore, fairness is described as the absence of bias in AI systems which is

challenging to achieve. Several ways are described to approach fairness. This includes group fairness, in which different groups are treated equally. Individual fairness, in which individuals are treated similarly regardless of their groups. Transparency, making the AI system's decision-making public. Lastly, intersectionality, in which the way different demographics or identities interact are considered.

In relation, the study by Yang et al. (2024) investigated the causes of biases in AI models in the data pipeline from data collection to model development. They have pinpointed various sources of biases in AI systems that can lead to biased datasets and machine learning systems. This includes historic bias, which exists because of past societal prejudices or inequalities. Representation bias, which happens when some groups are underrepresented or overrepresented in the dataset. Aggregation bias, which exists when data from different groups are combined even though there's a difference between them. Population bias, which exists when the population used to train the model is not the same where it is applied. Lastly, measurement bias, which occurs when the tools or methods used to collect data are inaccurate.

Furthermore, Hanna et al. (2025) highlighted that achieving fairness requires bias auditing and mitigation across the entire data pipeline. In preprocessing, raw data should be checked for imbalance and historical bias before feeding it into a machine learning model. They emphasized that a model is only truly fair if its accuracy and reliability does not degrade when applied to underrepresented groups. Overall, the theory of Ferrara, Yang et al., and Hanna

et al. suggests that fairness is achieved when the different kinds of bias found in the data preprocessing pipeline are addressed.

Hybrid Metaheuristic Algorithm

Somvanshi et al. (2025) recommend hybridization to address performance gaps of metaheuristic algorithms. On one hand, foundational algorithms such as “Differential Evolution” are considered mathematically strong in exploration but suffer from slow speed. On the other hand, newer algorithms like “Whale Optimization Algorithm” are effective in exploitation but are prone to local-optima. By merging the robust exploration capabilities of DE and strong exploitation efficiency of WOA, its individual weaknesses can be overcome.

Fitness Function (Fairness)

The fitness function framework serves as a mathematical evaluation tool combining Statistical Parity Difference (SPD) and Disparate Impact (DI) to measure fairness of protected attributes like race, sex, and age. These metrics compare a privileged group to an unprivileged group. These groups are referred to as the intersectional demographic with either the highest or lowest rate of favorable outcomes (Castelnovo et al., 2022). Fitness function is calculated as:

$$f(X) = |StatisticalParityDifference| + |1 - DisparateImpact|$$

On one hand, SPD will measure the absolute difference between group outcomes to quantify exactly how large the bias gap is. On the other hand, DI will

measure the ratio of favorable outcomes that different groups encounter (Osborne & Usani, 2026). The formula for SPD and DI is as follows:

$$SPD = P(\hat{Y} = 1|A = u) - P(\hat{Y} = 1|A = p)$$

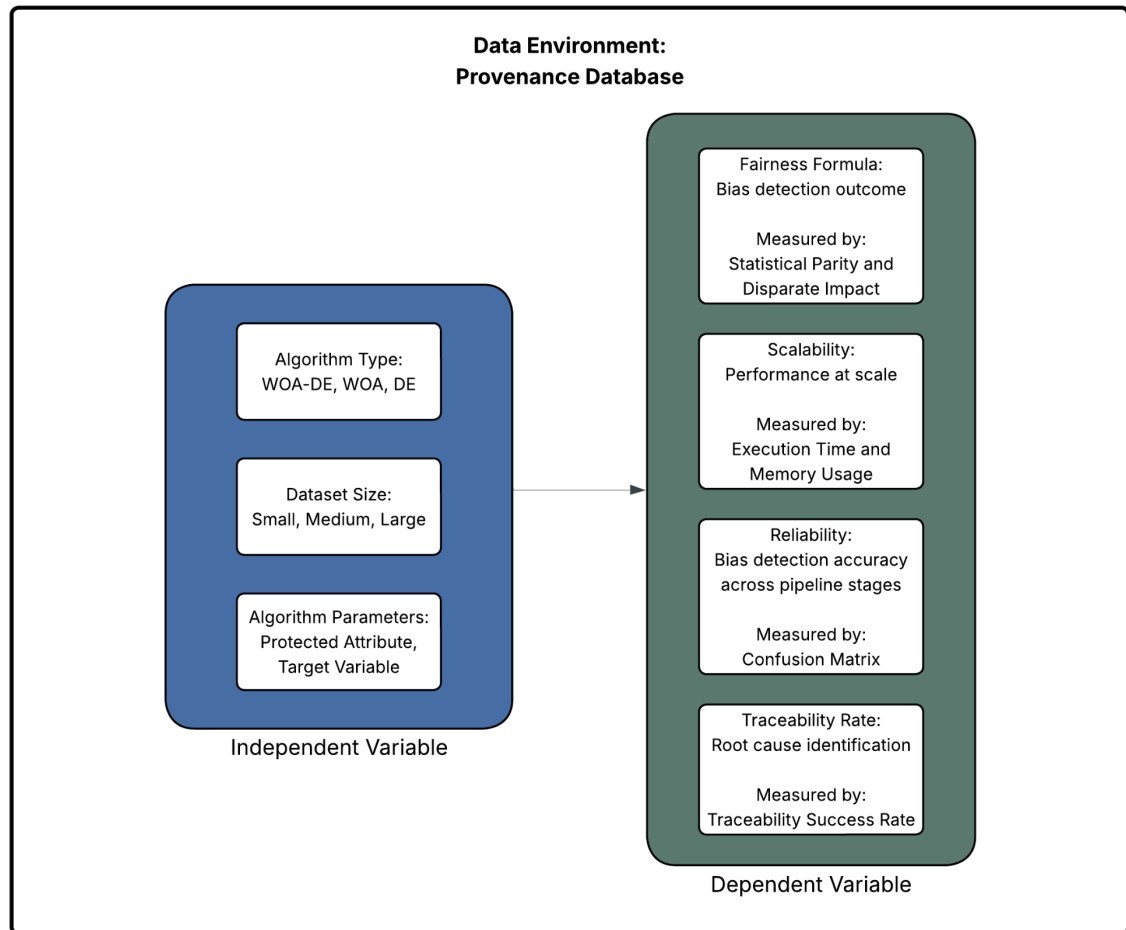
$$DI = \frac{P(\hat{Y} = 1|A = u)}{P(\hat{Y} = 1|A = p)}$$

In the given formula, 'A = u' represents the unprivileged group, 'A = p' represents the privileged group, 'P(Y=1 | A = u)' represents the probability of a positive prediction for the unprivileged group, and 'P(Y=1 | A = p)' represents the probability of a positive prediction for the privileged group.

Conceptual Framework

The IV-DV model system was used to present the independent and dependent variable of the research to be done. This provides a structured approach for detecting demographic bias within data pipelines. The system operates by transforming raw, unverified data into audited, traceable datasets through a hybrid algorithmic approach.

Figure 3: Conceptual Framework



1. Independent Variable

This includes the components that are deliberately controlled and manipulated by the researchers.

- **Algorithm Type:** The study tests three algorithmic configurations, a hybrid approach (WOA-DE), and its two individual counterparts (WOA

and DE), to compare their effectiveness in detecting bias within data pipelines.

- **Dataset Size:** The system is tested across three dataset sizes (small, medium, and large) to evaluate how each algorithm performs under varying data volumes. This allows for the evaluation of the system's scalability.
- **Algorithm Parameters:** The user defines the demographic variables to be monitored for ethical violations and the binary outcome variable required for computing fairness metrics.

The independent variables establish the setup required to solve the problem. During preparation, a known bias will serve as a target (ground truth) for each algorithm to detect. The algorithm parameters provide the necessary initial inputs required for the system to execute the audit.

2. Dependent Variable

This includes the components that are observed and measured as a result of the manipulation of the independent variable.

- **Fairness Formula(Or Fitness Function):** This component evaluates bias detection outcomes by determining whether the system produces equitable results across demographic groups within the data pipeline. It is measured by Statistical Parity and Disparate Impact.
- **Scalability:** This component evaluates the computational performance of the system as it processes varying dataset sizes. It is measured by Execution Time and Memory usage.

- **Reliability:** This component evaluates how consistently and accurately the algorithm identifies bias instances across the different stages of the data pipeline. It is measured by a Confusion Matrix.
- **Traceability Rate:** This component evaluates the algorithm's ability to identify the root cause of detected bias by measuring the percentage of trials in which it successfully traces the highest bias point to its origin within the data pipeline. It is measured by the Traceability Success Rate.

The dependent variables serve as how the system's overall effectiveness is measured. The goal is to solve the problem of silent bias that is difficult to track. The system transforms raw, uncleaned data into audited, traceable logs. The fairness formula serves as the algorithm's brain, guiding the search through the metadata logs. Meanwhile, scalability, reliability and the traceability rate are the key metrics to evaluate how well the system performs.

Statement of the Problem

The intent of the study is to identify and detect bias using metadata provenance and WOA-DE Algorithm to determine the exact cause of bias. Specifically, this study aims to answer the following questions:

1. What is the performance of the hybrid WOA-DE algorithm compared to standalone WOA and DE algorithm in terms of:
 - 1.1. Fairness
 - 1.2. Scalability
 - 1.3. Reliability

2. What is the traceability success rate of the metadata provenance system in correctly identifying the root cause of the detected bias?

Hypothesis

H₀: The hybrid WOA-DE algorithm fails to outperform the standalone WOA and DE algorithms in terms of fairness, scalability, and reliability.

H₁: The hybrid WOA-DE algorithm outperforms the standalone WOA and DE algorithms in terms of fairness, scalability, and reliability.

Scope and Limitation of the Study

The scope of this system is restricted to auditing bias within tabular datasets that contain demographic attributes. To achieve this, the study focuses specifically on implementing an algorithmic hybrid WOA-DE approach during the data preprocessing phase. The system is designed to evaluate specific dataset parameters and does not extend to other forms of data or bias audit outside of this designated hybrid approach.

While the proposed system provides a method for auditing bias, the researcher acknowledges several limitations regarding its data requirements, system capabilities, and methodological stability:

1. The system is designed to work on a dataset containing a binary target variable.
2. The dataset must first pass through the preprocessing pipeline to generate the metadata provenance logs before the audit can occur.

3. The hybrid algorithm is not intended to operate directly on the raw dataset, but rather to navigate across the metadata provenance logs.
4. The system relies on established fairness formulas to trigger the algorithm.
5. The system's traceability is restricted to recording the automated edits made by the pipeline itself. It lacks the capability to track or record any manual edits or interventions made by a human operator.
6. The system is strictly a diagnostic auditing tool for bias detection. While it provides flagging of the specific transformation causing an ethical violation, it does not include features for the automatic mitigation, correction, or cleaning of the identified bias.

Significance of the Study

The purpose of the study is to provide a new way to handle fairness in automated systems that use demographic data. This study will also benefit:

Data Analyst, Data Scientists, and Data Engineers. Since the system records an edit history or metadata provenance, it creates a chronological paper trail of every change made during the data cleaning process. This saves time because a biased result can be traced back to its specific origin in the pipeline, which makes it much easier to fix the logic compared to checking everything manually.

AI Ethics Auditors and Organizations. The audit process becomes more efficient and scalable when dealing with large datasets. It ensures that datasets used to train models are fair. This also ensures that organizations can maintain transparency and accountability by having a traceability log for their data.

General Public. The system prevents ethical violations and ensures that people are treated fairly by monitoring protected attributes such as age, sex, and race.

Future Researchers. This study serves as a guide for using metaheuristic optimization and metadata to solve computing problems related to AI ethics. It fills a gap in current research by focusing on the active provenance of data instead of just looking at the final output.

Definition of Terms

The following terms have been used operationally in this study:

AI Ethics. Principles that ensure AI aligns with human values and societal well-being (Karl, 2024). In this study, it involves auditing bias in preprocessing pipelines to ensure AI systems are transparent, fair, accountable, and secure.

Bias. An unfair treatment to a group that leads to unfair outcomes (Veenendal, 2024). In this study, it is the prejudice in favor of or against one thing, person, or group that causes machine learning models to produce unfair, inaccurate, or prejudiced outcomes.

Bias Auditing. A systematic evaluation to detect, evaluate, and mitigate biases to determine if a system shows unfair treatment toward groups (Holistic AI, n.d.). In this study, bias auditing will take place in the data preprocessing pipeline.

Bias Detection. The process of examining AI models and datasets to identify unfair treatment among different demographic groups to promote fairness and transparency (OneTrust, n.d.). In this study, the root cause of biases introduced in the data preprocessing pipeline will be detected.

Bias Mitigation. The techniques employed to detect and minimize bias in data, algorithms, and decision-making systems (arc42, n.d.). In this study, a recommendation-based mitigation is delivered in the output as human oversight is necessary before applying mitigation techniques.

Convergence. The point where all solutions become very similar and no further improvements are observed (Alorf, 2023). In this study, convergence happens when the hybrid WOA-DE settles on the best possible solution and stops making improvements.

Data Pipeline. Series of processes that transfer data from one or more sources to a target destination, often transforming it during the process (Johnson, 2026). In this study, the tool audits bias in the preprocessing stage of the data pipeline.

Data Preprocessing. The process of cleaning and transforming raw data to make it suitable for analysis and model development (GeeksforGeeks, 2026). In this study, bias introduced from the data preprocessing stage will be detected and pinpointed.

Differential Evolution (DE). An algorithm that iteratively improves potential solutions using mutation and crossover to fine-tune the search (Shaji, 2021). In this study, DE acts as the local exploiter and pinpoints the exact cause of bias.

Demographic. The statistical traits of human population, such as age, race, income which is used to analyze markets (Merriam-Webster, n.d.). In this study, demographic bias is the type of bias being detected and audited in the data preprocessing pipeline.

Exploitation. The process of focusing the search within promising areas to find the best possible solution (Alorf, 2023). In this study, the DE will act as a local exploiter to pinpoint the exact cause of bias.

Exploration. The process of broadly searching for potentially promising areas (Alorf, 2023). In this study, the WOA will act as a global explorer to search for groups of bias.

Fairness. Handling data responsibly and making decisions that do not cause discrimination to any individual or group (Veenendal, 2024). In this study, fairness is used to evaluate the tool's performance by measuring how fair the cleaned dataset is using the Fitness Function.

Global-Optimum. Best solution in the entire search space. (KylIn, n.d.). In this study, the WOA-DE is used to search for the global optima.

Intersectional Bias. The intersection of multiple protected attributes which expose hidden bias that can't be detected by a single attribute alone (Borenstein, 2023). In this study, multiple protected attributes can be selected as an input to detect intersectional bias.

Local-Optimum. The best solution within a limited area of the search space. (KylIn, n.d.). Algorithms can sometimes get trapped here, thinking it's the most optimal solution. In this study, the WOA, who are prone to get stuck in local optima, is combined with DE to overcome this weakness.

Metadata Provenance. The history, origin, or transformation of data that makes research transparent (EDI, n.d.). In this study, the data preprocessing will be wrapped with Python decorator to get the metadata provenance of a record.

Metaheuristic Optimization. A high-level strategy used to explore complex search spaces and solve complex problems (Nature Index, n.d.). In this study, it will be used to solve and optimize the search for the metadata logs generated to find the optimal solutions.

Protected Attribute. Personal characteristics that are legally shielded from discrimination such as age, color, and disability (DC Office of Human Rights, n.d.). In this study, this refers to the columns or attributes to audit if it has bias.

Record. In databases, it is the row or collection of related fields (columns) that store information about one entity (CS Unfolds, 2025). In this study, the scalability is measured by increasing the number of records in the dataset.

Reliability. Produces precise and dependable results when used on different data sets over time (Pecan, n.d.). In this study, reliability is used to evaluate the tool's performance by measuring if the detected bias is really a bias or not using F-1 Score.

Scalability. The capacity of a system to support growing and larger workloads (Kaur, 2023). In this study, scalability is used to evaluate the tool's performance by measuring its time and space complexity on increasing numbers of records from the dataset.

Silent Bias. Bias that begins in unexamined data pipelines and propagates downstream in the pipeline, which makes it harder to detect and correct (Trivedi, 2025). In this study, the data preprocessing pipeline is monitored using a bias auditing tool to avoid silent bias.

Target Variable. The variable the model is trying to estimate based on input data. (Smolic, 2024). In this study, the target variable defines the outcome variable (e.g. Loan approval, Hiring decision).

Traceability. Complete record of a data's journey as it moves, transforms, and is used from origin to final use (MacDonald, 2024). In this study, traceability refers to correctly identifying the root cause of the detected bias through metadata provenance.

Training Data. Data used to teach AI models to identify patterns, make decisions, or predict outcomes (Hass, 2025). In this study, biases in the data preprocessing pipeline are audited to avoid training data used in AI to harm specific demographic groups.

Whale Optimization Algorithm (WOA). An algorithm mimics the spiral bubble-net hunting behavior of humpback whales (Gomede, 2023). In this study, WOA acts as the global explorer and searches through the massive metadata logs for potential clusters of bias.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents a review of related literature and studies relevant to the present research. It discusses theories, concepts, and findings from previous studies that are closely related to the variables of the study and provides a foundation for the research.

Dataset Bias as the Ethical Foundation of AI Systems

The widespread deployment of machine learning systems across critical sectors, including healthcare, finance, education, and criminal justice, has transformed dataset quality from a purely technical concern into a matter of ethical and social consequence. Biased training data can harm entire demographic groups even when models appear statistically accurate. This position is explicitly articulated in *Bias in Artificial Intelligence: Smart Solutions for Detection, Mitigation, and Ethical Strategies in Real-world Applications* by Samala and Rawas (2025), who define bias as a structural phenomenon embedded within datasets, algorithms, and social systems.

Types of Bias in AI Systems

Samala and Rawas (2025) identified three dominant forms of bias: gender, racial, and socio-economic, which originate from biased training data, algorithmic design choices, and historically rooted inequalities. Their work emphasizes that bias is not introduced at the modeling stage alone but enters the AI pipeline during data collection, curation, and labeling.

This structural framing is supported by *A Survey on Bias and Fairness in Machine Learning* by Mehrabi et al. (2021), which introduces a formal taxonomy of dataset bias. The authors classify bias into representation bias, measurement bias, and historical bias. Representation bias arises when certain populations are underrepresented, measurement bias occurs when data collection tools perform inconsistently across groups, and historical bias reflects the continuation of social inequalities preserved in legacy datasets. These forms of bias often interact, making post hoc correction difficult.

Further reinforcing the socio-technical origins of dataset bias, Suddala (2021), in *Exploring the Ethical Implications of Biased Datasets on Decision-Making*, explains that bias emerges through historical inequities, sampling bias, measurement bias, and label bias. Label bias, in particular, arises from human subjectivity during annotation, embedding ethical distortions directly into ground-truth labels.

Most Used Attributes in Demographic Datasets

Demographic attributes are fundamental in detecting and evaluating bias in machine learning systems as they enable researchers to compare model performance across different groups. Based on the findings of multiple studies, gender, age, and race/ethnicity appear across all studies regardless of study domain or focus. These attributes function as the standard demographic attribute in fairness and bias research due to its availability and ease of annotation (Andrus et al. (2022); Cascone et al. (2026); Colacci et al. (2025); Dominguez-Catena et al. (2025); Majumdar et al. (2025); Sufian et al. (2025)).

Beyond the three, several studies also use socioeconomic status, education level, and nationality in demographic. (Colacci et al. (2025); Dominguez-Catena et al. (2025); Majumdar et al.)). However, they are not common across all domains of machine learning and are only present when needed by the research focus.

Intersectional Bias

A system may appear fair across broad demographic categories (e.g. race, age, gender) but still unfair to intersecting subgroups (e.g. race x age, race x gender, age x gender) (Stewart, 2022). This concept is called intersectional bias, where discrimination or bias emerges from the intersection of multiple protected attributes rather than a single attribute alone.

Boxer et al. (2023) introduced the Conditional Bias Scan (CBS) method, which automatically searches through many combinations of demographic subgroups to detect hidden biases. The method was evaluated using the COMPAS dataset on a real AI system. Likewise, Suk & Han (2024), introduced a method called Interactive Differential Algorithmic Functioning (Interactive DAF) approach, which compares how algorithms treat individuals with similar characteristics across different intersectional groups.

Both studies discovered hidden intersectional biases that traditional fairness methods miss. These findings suggest that the bias auditing system also needs to look for intersectional bias, rather than relying only on single-attribute evaluations.

In addition, Borenstein et al. (2023) examined how racism and sexism appeared in historical newspapers from the colonial Caribbean. The study measured gender bias, racial bias, and their intersection. Findings show both gender and racial bias are deeply connected and that intersectional groups have experienced unique stereotypes that cannot be explained by gender or racial bias alone. This supports the idea that there are hidden biases that only appear when multiple protected attributes intersect.

Technical and Ethical Impacts of Biased Training Data

Dataset bias produces two distinct but interrelated harms: technical degradation and ethical injustice. Technical bias refers to failures in generalization, while ethical bias refers to systematically unfair outcomes for specific groups, even when overall accuracy remains high.

The technical consequences of biased datasets are empirically demonstrated in *Tackling Dataset Bias With an Automated Collection of Real-World Samples* by Sevetlidis et al. (2022). Results show that models trained on single-source datasets exhibited accuracy drops from approximately 70% to 48% when evaluated on external datasets. In contrast, models trained on multi-source datasets maintained stable performance across environments. This finding confirms that diversity in data sourcing is not merely ethically desirable but functionally necessary.

The ethical implications of biased datasets are emphasized in *Human-Centered Design to Address Biases in Artificial Intelligence* by Chen et al. (2023). The authors demonstrate that conventional evaluation practices often

prioritize aggregate performance metrics, which conceal subgroup disparities. As a result, marginalized populations are disproportionately harmed even when systems appear statistically reliable.

These harms extend into organizational and industrial contexts. In *Biases in AI-Supported Industry 4.0 Research: A Systematic Review, Taxonomy, and Mitigation Strategies*, Arévalo-Royo et al. (2025) show that dataset bias undermines not only predictive outcomes but also stakeholder trust, decision-making quality, and system legitimacy. Their phase-oriented framework reveals how bias propagates across the AI lifecycle.

Bias Propagation in Data Preprocessing Pipelines

The data preprocessing phase is a critical point of failure where ethical distortions are often introduced before an algorithm even begins training. According to Victor (2022), the specific selection of preprocessing techniques, such as how categorical variables are encoded or how features are scaled, can directly determine whether a model output is fair or biased. This confirms that simply auditing the final results of a system is insufficient because the intermediate transformations themselves dictate the ethical outcome. This problem is often compounded by cascading bias, a phenomenon discussed by Hancox-Li (2020) where small, seemingly insignificant errors in early data cleaning stages grow into major discriminatory patterns by the time the data reaches the output stage. These findings establish that without a structured way to trace the history or lineage of these edits, identifying the root cause of an unfair result is nearly impossible.

Black et al. (2023) argued that machine learning fairness is currently too theoretical and requires real-world practice across the entire ML pipeline. It was highlighted that current or traditional methods, such as adjusting models are no longer sufficient. They also emphasized that fairness problems must be addressed end-to-end, from problem definition to evaluation, in order to address the root cause of bias and fix it there.

Bias in Data Pipelines

A study by Ho & Beyan (2020) explained that bias can enter at every stage of the data pipeline. They emphasized that bias in data science is not just a model problem, but a pipeline problem. Six key stages were highlighted where bias can form which includes: Problem definition, where bias may form based on how the problem is posed. Data collection, which may exclude certain groups. Data Preparation, which may include subjective decisions on how to handle the cleaning, labeling, and preprocessing procedures. Modeling, where models learn patterns that reflect social biases. Evaluation, in which test data share the same biases as training data. Lastly, deployment and feedback, in which feedback from users may worsen the bias from an existing system over time. The paper suggests that being transparent about decisions and transformations, using diverse and representative datasets, evaluating models with fairness-aware metrics, and monitoring the system after deployment is the key to avoid biases.

Human Factor

Beyond the technical operations, the human factor remains a significant and often undocumented source of bias. Subjective decisions regarding what

counts as an outlier or how to handle missing values can embed human prejudice directly into the cleaning scripts. Furthermore, the technical standards used in feature engineering can unintentionally skew results for specific demographic groups, such as age or race, if the transformations are not monitored for ethical parity (Koukaras & Tjortjis, 2025). Because traditional software debugging tools are designed to find functional code errors rather than ethical ones, there is a clear need for fairness debugging frameworks like Gopher to specifically target these preprocessing stages (ProQuest, 2023). These studies collectively justify the need for an automated system that can monitor both the technical and human-driven changes within a pipeline in real-time.

Bias Injection

The literature identifies two primary pathways for the systematic injection of bias during the data preprocessing stage: Label Bias and Selection Bias. Mehrabi et al. (2021) define Label or Measurement Bias as the systematic shifting of target outcomes for specific protected groups. For example, flipping positive labels to negative for a minority subgroup to simulate historical human prejudice. Complementing this, Jesus et al. (2022) and Mehrabi et al. (2021) describe Selection or Representation Bias as the manipulation of subgroup samples. This is achieved by over-representing a majority group in positive outcomes or under-sampling a minority group, thereby forcing the machine learning model to learn spurious correlations between demographic traits and the target label. Together, these methods allow for a comprehensive simulation of the types of bias your auditing tool is designed to detect.

Metadata Provenance and Lineage as Infrastructure for Ethical AI

While biased outcomes manifest at the model level, their origins lie in the hidden decisions embedded within data pipelines. Identifying these origins requires more than inspecting samples. It requires structured documentation of data origins, transformations, and contextual decisions. This is the role of metadata provenance and lineage.

In *The Impact of Modern AI in Metadata Management*, Yang et al. (2025) argue that metadata management has shifted from manual, rule-based systems to AI-driven automation. Traditional approaches, they note, are labor-intensive, inconsistent, and incapable of scaling with data complexity. By integrating NLP and machine learning, metadata processes such as data discovery, cataloging, and governance can be automated, enhancing transparency and accountability.

Metadata as an Ethical Instrument

The ethical role of metadata is empirically demonstrated by Boyd (2021) in *Exploring the Ethical Implications of Biased Datasets on Decision-Making*. When engineers were provided with detailed metadata, including sourcing criteria, demographic composition, and preprocessing steps, they identified ethical risks earlier and more reliably. This confirms that metadata is not merely descriptive, it is diagnostic.

Di Benedetto (2024), in *Using Metadata: The Ethical Problems*, formalizes this ethical function by defining five quality dimensions: accuracy, accessibility, coherence, completeness, and timeliness. He further argues that usability must

be ensured at three levels: writer, reader, and repository, so that ethical assessments can be operationalized rather than merely documented.

Provenance for Explainability and Trust

In *Machine Learning Model Explainability Supported by Data Explainability*, de Oliveira et al. (2024) argue that model-level explainability is insufficient without data-level transparency. By linking model outputs to preprocessing transformations, provenance enables features to be traced back to raw sources. Without this linkage, explainability tools become opaque abstractions.

This governance role of provenance is further developed in *Data Lineage and Impact Analysis* by Karanam (2022), which positions lineage as the only reliable mechanism for root cause analysis in complex data ecosystems. Lineage enables proactive compliance, impact assessment, and change management.

Provenance in Regulated and High-Stakes Domains

Ahmed et al. (2023), in *Data Provenance in Healthcare: Approaches, Challenges, and Future Directions*, show that provenance bridges regulatory requirements with technical enforcement. However, they note that most existing approaches remain conceptual and fail to classify provenance records based on sensitivity.

This regulatory framing is echoed in *Building Ethical and Trustworthy AI Systems through Effective Metadata Management* by Sehgal and Bafna (2025), which emphasizes that emerging regulations such as the EU AI Act require

detailed documentation of system lifecycles, dataset sources, and preprocessing steps.

Scalability and Automation of Lineage

In *Hofmann’s Tracer: A Machine Learning Approach to Data Lineage* (2020), automated lineage reconstruction is proposed for environments where manual documentation is infeasible. While effective for primary key detection, the system struggles with complex relational structures.

Paul (2024–2025), in *Data Lineage and Provenance Tracking for Multimodal ERP Analytics at Scale*, argues that traditional tools fail in heterogeneous environments and proposes unified metadata schemas and knowledge graphs.

The systemic consequences of undocumented lineage are exposed in *The Data Provenance Initiative* by Longpre et al. (2023), which identifies widespread license ambiguity, attribution errors, and geographic bias across 1,800 datasets.

Ground Truth

Establishing a reliable ground truth in algorithmic fairness is traditionally difficult due to the fairness-accuracy trade-off inherent in existing data. Current research, specifically Defrance et al. (2025), utilizes the Fair World Framework. This methodology assumes a fair underlying state by utilizing a clean baseline dataset where demographic parity is maintained. Any bias subsequently detected

after injection can then be definitively attributed to the researcher intervention rather than random statistical noise.

To enhance this, Alu et al. (2026) emphasize the role of metadata provenance and standardized logs to record data pipeline transformations. This creates a traceable record that distinguishes true bias detection from variation, providing a rigorous benchmark for auditing precision.

Challenges in Bias Detection

Bias detection constitutes one of the most critical yet unresolved challenges in the development of ethical artificial intelligence systems. While prior themes establish that bias is structurally embedded within datasets and that metadata provenance enables its traceability, this theme focuses exclusively on the difficulties of identifying bias reliably, consistently, and at scale once it is already embedded in AI systems. The literature demonstrates that existing detection approaches remain methodologically unstable, computationally demanding, and weakly integrated into governance infrastructures.

Bias detection methods are designed to act as diagnostic instruments that reveal disparities in representation, association, and performance across demographic groups. These methods range from static word-embedding association tests to contextualized and dynamic evaluation frameworks. However, despite their widespread use, these tools often fail to produce consistent and reproducible results, undermining their practical usefulness.

In *Mind Your Bias: A Critical Review of Bias Detection Methods for Contextual Language Models*, Husse and Spitz (2022) conducted a systematic evaluation of widely used bias detection tools such as SEAT, CEAT, and LPBS. Their study revealed that bias measurements are highly sensitive to experimental configurations. Minor methodological changes, such as switching from personal names to group identifiers, modifying contextual prompts, or altering output encodings, can drastically alter bias scores for the same model. In several cases, these changes shifted results from statistically negligible to strongly significant.

Husse and Spitz (2022) further identify inconsistencies in code implementations and reporting practices across different studies. These inconsistencies cause reproducibility failures, where bias scores often reflect tool configuration rather than genuine model behavior. Consequently, current detection pipelines lack stability and standardization, making them unreliable for real-world deployment. This instability prevents bias detection from functioning as a dependable ethical safeguard.

Beyond technical fragility, governance-related limitations further complicate bias detection. In *Mitigating Bias and Advocating for Data Sovereignty: The Role of Metadata and Paradata in Ethical AI-Driven Information Systems*, Carrasco (2025) argues that detection mechanisms cannot function meaningfully in isolation from documentation, transparency, and accountability frameworks. She distinguishes between metadata, which documents dataset attributes, and paradata, which records the procedural and decision-making processes that shape datasets. Without this dual-layer documentation, detection becomes fragmented, unverifiable, and difficult to enforce.

Carrasco (2025) further emphasizes that bias detection must align with sovereignty-oriented and regulatory principles, such as CARE (Collective Benefit, Authority to Control, Responsibility, and Ethics). This framing establishes that detection is not merely a technical operation but a governance obligation. When detection mechanisms are disconnected from policy frameworks and documentation systems, fairness assessments become symbolic rather than actionable.

The literature demonstrates that current bias detection approaches are methodologically unstable, computationally inefficient, and weakly embedded within governance infrastructures. These limitations prevent detection systems from operating as reliable ethical safeguards. As a result, bias detection requires scalable, standardized, and infrastructure-level solutions capable of functioning consistently in complex, real-world AI environments.

Bias Detection Tools Gap

Several studies have proposed and introduced different frameworks, algorithms, and methods in detecting biases in datasets whereas each has its own gaps and limitations that needs to be addressed and implemented.

Oveh and Isitor (2025) introduced a Bias Audit Framework (BAF) that detects and fixes algorithmic bias during the early stages of AI development. The framework uses four components: data bias assessment, model bias evaluation, developer training, and continuous monitoring. Using a healthcare dataset, applying BAF improved the fairness while making sure that the model is just as accurate. The study highlights that fairness can be improved without sacrificing accuracy. However, the framework is not yet fully practical for real-world use as it

faces a fairness vs accuracy trade-off and needs continuous monitoring. Furthermore, bias can still exist and reappear at different stages of the pipeline.

Li et al. (2025) proposed a method called “Decoupled Confident Learning” (DeCoLe) to identify mislabeled data in datasets while considering bias across different groups. Unlike traditional mislabeling detection, DeCoLe operates on the assumption that labeling mistakes depend on both class and group. Results show that DeCoLe performs better than existing methods and that the improvement is especially strong when there is bias across groups. However, the method is limited to datasets that have group labels, struggles with underrepresented groups, and is harder to scale.

Bao and Barzilay (2022) proposed the “Learning to Split (LS)” algorithm that detects hidden bias by repeatedly splitting the dataset into train and test sets, training a model, and adjusting the split. Results show that it successfully found bias and challenging splits across different datasets. However, the LS algorithm exhibits notable limitations, such as the inability to explain what the bias is, is heavily model-dependent, and is computationally expensive.

Known organizations such as Microsoft and IBM have developed bias detection and fairness assessment tools. On one side, Fairlearn is an open-source tool developed by Bird et al. (2020) at Microsoft Research that measures and mitigates fairness issues in machine learning. It includes visualization tools to compare model performance across different groups to balance fairness and accuracy using metrics such as demographic parity and equalized odds. On the other hand, AI Fairness 360 (AIF360) is an open-source

toolkit developed by Bellamy et al. (2018) at IBM Research that measures and mitigates bias in machine learning systems. It uses statistical fairness measures and protected attributes to evaluate bias. It also provides mitigation techniques that can be applied before, during, or after model training. Despite the strength of both Fairlearn and AIF360, both tools have limited built-in support for intersectional bias. They are also not designed for traceability and root-cause analysis, which are gaps that this study aims to address.

Controlled Bias Injection in Fairness Auditing

The literature establishes that injecting controlled bias is a fundamental requirement for validating audit frameworks. Jesus et al. (2022) argue that because real-world datasets contain hidden or unknown levels of historical bias, they are unsuitable for measuring the absolute sensitivity of an auditing tool. By injecting bias into a clean dataset, researchers create a robust test bed where the ground truth is known, allowing for the evaluation of whether an algorithm such as a hybrid optimization model can handle realistic, high-impact scenarios. Furthermore, Alu et al. (2026) highlight that controlled injection is the only viable method for determining detection thresholds, which is the precise point at which a subtle demographic skew becomes statistically significant enough for an auditor to flag.

Metaheuristic Optimization for Scalable Ethical Bias Detection

The detection of ethical bias within large-scale training datasets constitutes a computationally complex problem due to the high dimensionality, heterogeneity, and volume of metadata provenance. Exhaustive search methods become infeasible under

these conditions, as the search space grows exponentially with the number of dataset attributes, transformations, and lineage paths. As a result, bias detection in this context can be characterized as an NP-hard optimization problem. To address this, metaheuristic algorithms have been increasingly explored as scalable alternatives, as they balance global exploration and local exploitation to approximate optimal solutions within reasonable time constraints.

Metaheuristics operate by iteratively refining candidate solutions through stochastic search mechanisms. Global exploration allows the algorithm to traverse diverse regions of the search space, preventing early stagnation, while local exploitation refines promising solutions toward convergence. However, no single metaheuristic has been shown to consistently achieve an ideal balance between these two objectives across complex, high-dimensional problems.

This limitation is illustrated in *Energy-Aware Bag-of-Tasks Scheduling in the Cloud Computing System Using Hybrid Oppositional Differential Evolution-Enabled Whale Optimization Algorithm* by Chhabra et al. (2022). Their study demonstrates that standalone metaheuristics exhibit inherent performance trade-offs. The Whale Optimization Algorithm (WOA) is effective in local exploitation due to its bubble-net hunting strategy, which enables rapid convergence around promising solutions. However, this same behavior makes it prone to premature convergence in complex optimization landscapes. In contrast, Differential Evolution (DE) provides stronger global exploration through mutation and crossover mechanisms but often requires higher computational cost and exhibits slower convergence.

Chhabra et al. (2022) argue that this trade-off creates a performance threshold where no standalone algorithm consistently achieves both high solution diversity and fast convergence. As problem dimensionality increases, WOA tends to stagnate in local optima, while DE becomes computationally expensive for large-scale tasks. Although hybridization has been proposed to overcome these weaknesses, they note that many existing hybrid models fail to provide a systematic improvement across convergence speed, diversity, and efficiency simultaneously.

The structural limitations of standalone swarm-based optimization are further examined in *Multistrategy Improved Whale Optimization Algorithm and Its Application* by Liu and Zhang (2022). Their analysis confirms that WOA suffers from “insufficient global search ability and slow convergence speed” when applied to complex optimization problems. While WOA is conceptually simple and computationally lightweight, it cannot independently resolve the imbalance between exploration and exploitation, especially in rugged and high-dimensional landscapes.

These observations are consistent with findings in *Differential Evolution: A Review of More Than Two Decades of Research* by Bilal et al. (2020). The authors emphasize that as optimization problems grow in dimensionality and complexity, it becomes increasingly difficult to identify the nature of the problem. This uncertainty makes standalone metaheuristics vulnerable to selecting inappropriate search strategies, often resulting in stagnation or inefficient convergence. Their review highlights that no single strategy universally dominates across problem classes.

Bilal et al. (2020) also observe that certain optimized standalone variants of DE, such as JADE and SaDE, have achieved competitive performance in international

benchmarks, including IEEE CEC competitions. These variants rely on self-adaptive parameter tuning rather than full hybridization. They report that approximately 24% of documented performance improvements stem from modified mutation schemes alone. This suggests that, in some contexts, simpler adaptive strategies may rival complex hybrid models in terms of efficiency.

However, the problem of ethical bias detection in high-dimensional metadata provenance introduces unique challenges that may exceed the capabilities of standalone approaches. The search space in this context is not only large but also structurally complex, as bias can be embedded across multiple stages of a dataset's lifecycle. Random initialization strategies often fail to ensure sufficient coverage of this space, leading to incomplete or skewed detection results.

This motivates the need for enhanced optimization architectures that can traverse the entire data lineage while maintaining convergence efficiency. The performance limitations identified by Chhabra et al. (2022), Liu and Zhang (2022), and Bilal et al. (2020) collectively demonstrate that while standard metaheuristics provide a useful framework, they remain insufficient for large-scale, real-time ethical auditing tasks. These shortcomings form the basis for exploring hybridized approaches, which are examined in the succeeding theme.

Whale Optimization Algorithm (WOA) and Differential Evolution (DE) Hybridization

Hybridization has emerged as a dominant strategy in metaheuristic optimization to overcome the inherent limitations of standalone algorithms. Rather than relying on a single search behavior, hybrid models combine complementary strengths from multiple algorithms to enhance convergence speed, solution diversity, and robustness. In the

context of large-scale and high-dimensional problems, such as ethical bias detection in metadata provenance, hybridization is particularly valuable because it enables both wide search coverage and precise local refinement.

The limitations of standalone metaheuristics motivate this shift. As discussed in *Energy-Aware Bag-of-Tasks Scheduling in the Cloud Computing System Using Hybrid Oppositional Differential Evolution-Enabled Whale Optimization Algorithm* by Chhabra et al. (2022), WOA is prone to premature convergence due to its strong exploitation behavior, while DE, although effective in global exploration, incurs higher computational costs and slower convergence. These complementary weaknesses suggest that neither algorithm alone can consistently achieve optimal performance in complex optimization landscapes.

A concrete demonstration of hybridization is presented in *Multistrategy Improved Whale Optimization Algorithm and Its Application* by Liu and Zhang (2022). They proposed a hybrid model called the Differential Evolution Chaotic Whale Optimization Algorithm (DECWOA). This approach integrates DE operators into the WOA framework to improve global search ability while maintaining efficient local exploitation. Their method introduces “Sine chaos theory” to generate the initial population, increasing diversity and preventing early stagnation. Additionally, adaptive inertia weights are incorporated to dynamically balance exploration and exploitation throughout the optimization process.

Liu and Zhang (2022) reported that this hybrid structure allows the algorithm to avoid local optima more effectively than standalone variants. In complex engineering optimization tasks, the hybrid model achieved significantly higher accuracy, often

approaching near 100%, where standard WOA and DE struggled. These results demonstrate that hybridization can systematically address the structural weaknesses of individual algorithms rather than merely tuning their parameters.

This observation aligns with the broader trends identified in *Differential Evolution: A Review of More Than Two Decades of Research* by Bilal et al. (2020). Their review highlights that hybridization has become increasingly prevalent over the past two decades, particularly in complex and high-dimensional problem domains. While they acknowledge that some adaptive standalone variants of DE (such as JADE and SaDE) perform competitively, they also emphasize that hybrid models offer greater robustness when the nature of the optimization landscape is unknown or highly irregular.

In the context of ethical bias detection in metadata provenance, hybridization offers distinct advantages. Standard random initialization strategies often fail to distribute search agents uniformly across large and irregular search spaces. By incorporating chaotic mappings, such as the Sine chaos method discussed by Liu and Zhang (2022), a hybrid WOA–DE framework can ensure ergodic traversal of the solution space. This enables the algorithm to explore the full lifecycle of a dataset, increasing the likelihood of identifying hidden or subtle “audit gaps” where bias may be embedded.

Furthermore, the efficiency gains reported by Chhabra et al. (2022) suggest that hybridization can significantly reduce computational overhead. This is critical for making real-time ethical auditing feasible for massive training datasets. Rather than trading off between speed and accuracy, hybrid models aim to achieve both by design.

Recommendations for Mitigation

Bias mitigation is a technique employed to detect and minimize bias in data, algorithms, and decision-making systems (arc42, n.d.). Studies suggest that recommendation-based mitigation in bias auditing is more effective than automatic mitigation because fairness is highly context-dependent and may produce unintended consequences.

Chen et al. (2023) investigated 17 bias mitigation techniques if they work in practice. Their findings suggest that automatic mitigation can reduce model performance, inconsistently improve fairness, and worsen both in some cases. The study concludes that there is no optimal fairness mitigation technique, making human judgment necessary.

Furthermore, Krčo (2023) examined whether automatic bias mitigation creates new unfairness. The study shows that mitigation strategies designed to help one group can hurt another group, reduce model accuracy, and negatively affect the group it's supporting. The paper emphasizes the importance of deeper evaluation, transparency, and checking how mitigation affects real people. These findings suggest that human oversight is necessary before applying mitigation techniques.

Additionally, Pagano et al. (2022) highlighted that mitigation cannot be fully automated because fairness is context dependent. The effectiveness of bias mitigation depends on the dataset, protected attribute, task, and target variable (e.g. healthcare). This suggests that human-guided mitigation recommendations are more appropriate than automated corrections.

Synthesis

Dataset bias is not incidental. It is structural, persistent, and socially consequential. It can enter the data pipeline in the earlier stage, such as data preprocessing, making it not only a post-deployment error (Samala & Rawas (2025); Ho & Beyan, (2025)). Studies demonstrate that subjective human decisions and automated transformation silently introduce different kinds of bias in the early stage (Mehrabi, (2021); Suddala, (2021); Koukaras & Tjortjis (2025); ProQuest (2023)).

In bias detection, it's identified that age, race, and gender are the most used attributes in demographic dataset and usually used to determine bias. (Andrus et al. (2022); Cascone et al. (2026); Colacci et al. (2025); Dominguez-Catena et al. (2025); Majumdar et al. (2025); Sufian et al. (2025)). Hidden biases also emerge when multiple protected attributes intersect rather than a single attribute (Stewart (2022); Boxer et al. (2023); Suk & Han (2024); Borenstein (2023)). These biases propagate downstream, making them difficult to isolate, which is why addressing the root cause of bias requires early-stage intervention rather than post deployment correction (Victor (2022); Hancox-Li (2020); Black et al. (2023)).

While existing bias detection frameworks attempt to address bias, they are met with limitations that hinder their performance. This includes evaluating only in later stages of the data pipeline, inconsistencies in implementation, methodologically unstable, bias reappearing at different stages of the pipeline, struggles with underrepresented groups, computationally expensive, leaves intersectional biases undetected, and lacks built-in mechanism for real-time correction (Husse & Spitz (2022); Carrasco (2025); Oveh & Isitor (2025); Li et al. (2025); Bao & Barzilay (2022)).

To achieve traceability and root-cause analysis in biased datasets, metadata provenance is advocated as an essential governance infrastructure to enhance transparency and accountability (Yang et al. (2025); Boyd (2021); de Oliveira et al. (2024); Karanam (2022)). However, this generates massive, high-dimensional metadata logs that need to be searched for hidden biases.

Metaheuristic algorithm, like the hybridization of Whale Optimization Algorithm and Differential Evolution (WOA-DE) offer a scalable solution to navigate this complexity as they balance global exploration and local exploitation to approximate optimal solutions (Chhabra et al. (2022); Liu & Zhang (2022); Bilal (2020)).

To address the gaps mentioned, the study introduces a bias auditing tool that monitors the preprocessing stage of the data pipeline using metadata provenance. With the use of the WOA-DE algorithm, the cause of bias and intersectional bias can be pinpointed by going through the metadata recorded. This approach offers an opportunity to elevate the efficiency of a scalable, real-time auditing system within data pipelines.

CHAPTER 3

METHODOLOGY

This chapter contains the research methodology employed, the sources of data, instruments used, system architecture, data generation procedure, ethical considerations, including the data analysis and statistical treatment used in the study.

Research Design

The researchers will employ an experimental research design to achieve the thesis' aims, specifically focusing on the development and evaluation of the proposed bias audit system. According to Sreekumar (2024), experimental research is used to determine cause-and-effect relationship between variables and to test hypotheses. This approach is well-suited for this study because it involves measuring technical performance through data and testing the effectiveness of the hybrid WOA-DE algorithm. Since the main goal is to solve a computing problem related to ethics and data pipelines, the research is structured into several technical stages that contribute to the final solution.

This study also employs a quantitative methodology. Alex (2025) defines it as an approach that includes analyzing numerical data to produce measurable, comparable results using statistical tools. Through this structured approach, the researchers can effectively test the hypotheses regarding the tool's fairness, scalability, and traceability. By recording and analyzing the performance data generated during these stages, the study will provide a clear and evidence-based conclusion on the effectiveness of using metadata for bias detection.

Sources of Data

The data for this study will be sourced from the ACSIncome dataset, a publicly available dataset by Ding et al. (2022) which was derived from the American Community Survey (ACS) Public Use Microdata Sample (PUMS). The dataset, along with the accompanying Python package, is accessible through GitHub. It contains 1,664,500 records and 11 fields, with a binary target variable indicating whether an individual's annual income exceeds \$50,000.

Sampling Data

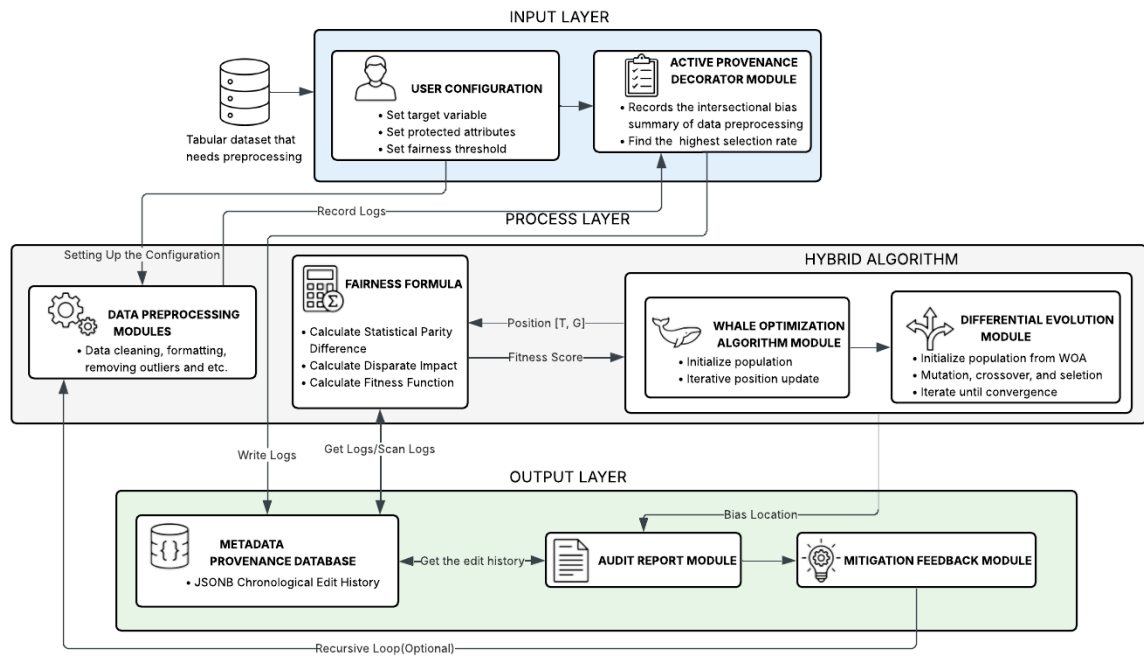
A purposive sampling technique is used to select a dataset that strictly aligns with the research objectives (Bisht, 2024). The dataset for this study must meet the following criteria:

1. The dataset must have at least three protected attributes in variables or columns.
2. The dataset must contain a binary target variable to allow the calculation of success rates for the fairness metrics.
3. The dataset must be publicly available or distributed under terms that permit legal use, reproduction, and analysis for research purposes.
4. The dataset must be in tabular form.

System Architecture

The proposed architecture represents an integrated, closed-loop framework designed to automate the lifecycle of bias detection and mitigation within data preprocessing pipelines. The system is organized into three functional layers: the Input Layer, the Process Layer, and the Output Layer.

Figure 4: System Architecture



The system architecture works as follows. First, a raw tabular file is accepted, where the user sets the target variable, protected attributes, and an allowable fairness threshold. This configuration is sent to the Active Provenance Decorator Module so that it knows which attributes and target binary variable to record. Once they are all configured, the dataset is sent to the Data Preprocessing module, where it undergoes cleaning, formatting, and outlier removal. The results of the data preprocessing are sent

back to the Active Provenance Decorator Module, which records the intersectional bias summary of the preprocessing and identifies the intersectional group with the highest selection rate to serve as the privileged baseline. During this recording process, a minimum sample size threshold ($N \geq 30$) is enforced to ensure statistical reliability and prevent the auditor from flagging random noise as systemic bias. These records are saved in the JSONB chronological edit history within the Metadata Provenance Database after the Fairness Formula module calculates the Statistical Parity Difference and Disparate Impact. The Hybrid Algorithm then scans these logs, where the Whale Optimization Algorithm module, which works as shown in Figure 5, sends specific coordinates (Position [P, T, G]) to request fitness scores from the fairness formula to scout for potential bias clusters. These findings are subsequently passed to the Differential Evolution module, which works as shown in Figure 6, to pinpoint the specific bias location. The audit findings are compiled by the Audit Report Module and passed directly to the Mitigation Feedback Module, which receives the localized bias coordinates from the algorithm to generate actionable insights. Serving as the final component of the system architecture, this module operates as a deterministic decision engine. It maps the detected bias hotspots directly to a predefined matrix of mitigation strategies. The recursive feedback loop is optional and occurs specifically when the user wants to have a report for every stage of the data pipeline.

Figure 5: Whale Optimization Algorithm

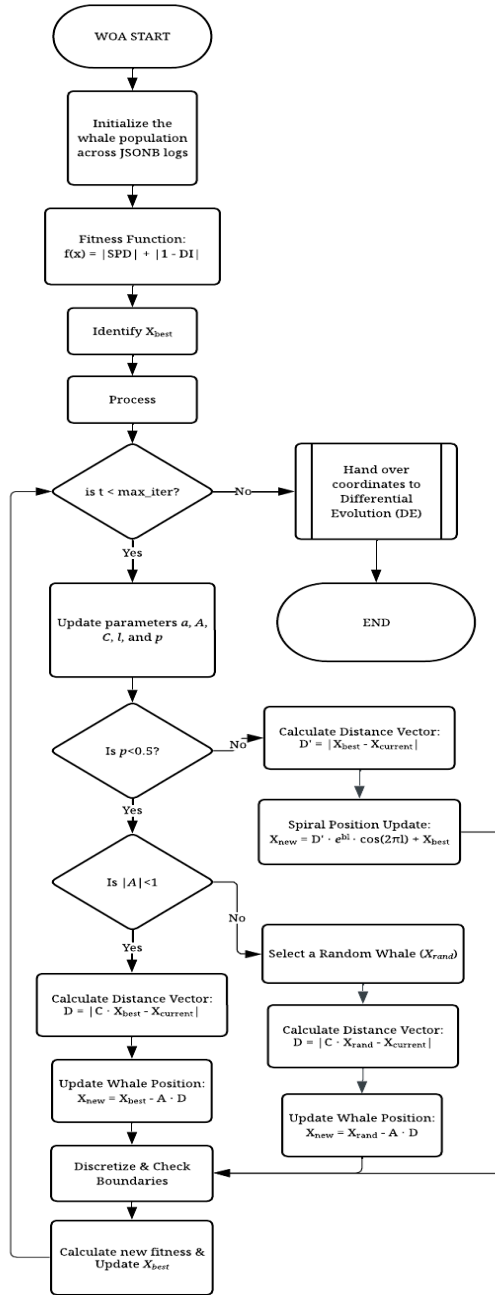
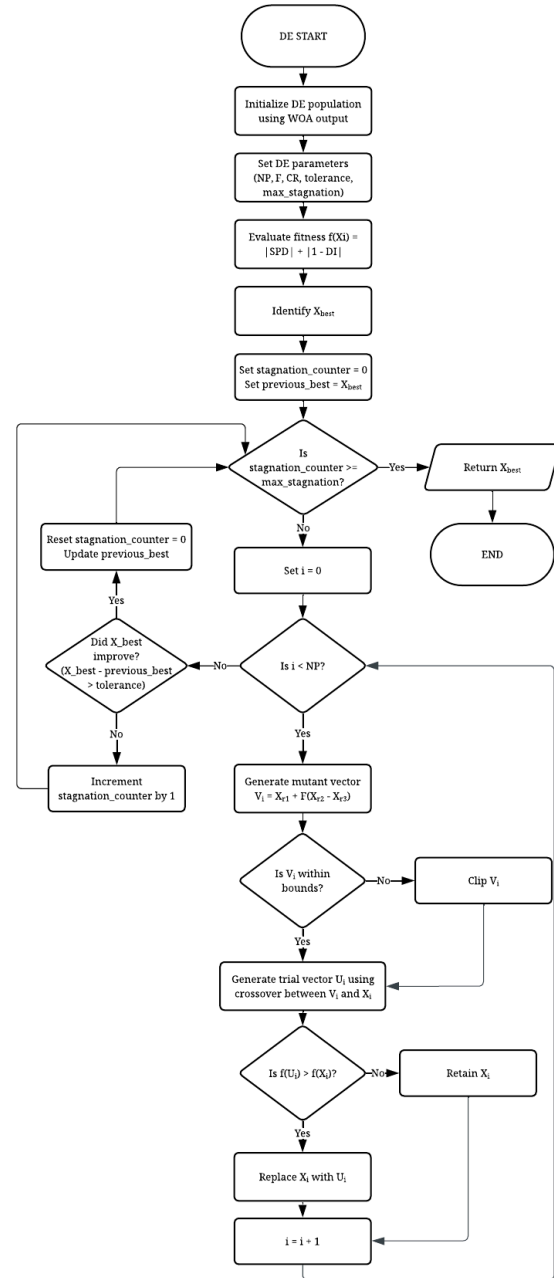


Figure 6: Differential Evolution



The flowchart above shows the hybridization process of the WOA-DE algorithm. The process will always start at WOA, the algorithm starts by initializing the population of whales scattered across the JSONB database, with their position vectors representing specific queries to different historical data transformation logs and demographic attributes. Next, the algorithm calculates the fitness of each search agent. This process involves retrieving the pre-calculated summary statistics from the metadata log and running the bias formula. The whale that discovers the highest bias score is designated as the current best search agent. After that, the algorithm enters a loop that runs until the maximum number of iterations is reached. Inside this loop, the algorithm updates its parameters for each search agent. The parameter a acts as a countdown, decreasing linearly from 2 to 0 to tighten the search over time, while p is a random probability that decides the whale's next movement into three distinct paths based on the values of p and the tuning coefficient A .

- **Spiral Bubble-Net Attack (Exploitation):** If $p \geq 0.5$, the algorithm moves to the spiral updating position. The whale use logarithmic spiral equation to search the immediate “neighborhood” wrapped closely around high bias metadata log found by X^* .
- **Encircling Prey (Exploitation):** If $p < 0.5$ and the absolute value of $A < 1$, the whales use the shrinking encircling mechanism. This forces the search agents to take small, precise steps to tighten their search net closer to the known ethical violation in the pipeline.
- **Search for Prey (Exploration):** If $p < 0.5$ and $A \geq 1$, the algorithm triggers the exploration phase. Instead of moving toward the known bias, the whales update their positions based on a completely random chosen whale from the population.

This crucial step makes our algorithm search globally across the entire JSONB database so it does not get trapped in a local optimum and miss other unfair pipeline steps.

After the movement phase, the algorithm regroups to check if any whale has gone beyond the search space boundaries and amends it. The fitness function is recalculated, and if a higher bias is found, the X^* is updated. Once the maximum iterations are completed, the WOA terminates and outputs the coordinates of the highest-bias metadata clusters and hands them over to the Differential Evolution.

The Differential Evolution algorithm receives the coordinates from the Whale Optimization Algorithm and uses them to initialize its population. This ensures that DE does not start from scratch but instead begins its search from a promising region of the solution space already identified by WOA. The algorithm then sets its necessary control parameters: NP, which is the population size, the scaling factor F, which controls the step size for exploration away from current solutions, the crossover rate CR, which is the probability determining how many traits are copied from the mutated vector compared to the original vector, tolerance, which is the minimum improvement required to consider a generation as progressive, and max_stagnation, which is the maximum number of generations allowed without meaningful improvement. After that, the fitness of each individual in the population is evaluated, and the initial highest-bias solution is identified.

The algorithm then enters its outer loop, which runs until the stagnation counter reaches the maximum stagnation limit. If the condition is met, the algorithm returns the best solution and terminates. Otherwise, it cycles through the entire population, processing each individual sequentially within the inner loop.

For each individual, DE performs three sequential operations:

- **Mutation:** A mutant vector is generated by selecting three distinct individuals at random from the current population. It calculates the difference between two of them, scales this difference using the scaling factor, and adds it to the third individual. This creates a new directional search point. If this new point falls outside the search space, it is clipped to the nearest boundary so it can remain a valid potential solution.
- **Crossover:** A trial vector is produced by combining the mutant vector and the original individual based on the crossover rate. Each coordinate of the trial vector is drawn from the mutant vector with a probability defined by the crossover rate, and the remaining coordinates are taken from the original individual, to ensure the offspring inherits traits from both vectors.
- **Selection:** The fitness of the trial vector is evaluated and compared against the current individual. If the trial vector yields a higher bias score, it replaces the current individual in the population. If not, the current individual is retained with no changes.

After all individuals in the population have been processed, the stagnation counter is either reset to zero if improvement was found or incremented by 1 if no meaningful improvement was observed. Then, the outer loop repeats until the stagnation limit is reached. Afterwards, the algorithm returns the refined coordinates of the highest-bias metadata clusters as its final output.

Research Instrument

This study uses a system developed by the researchers as the primary instrument for data generation and collection. The system processes raw datasets through data pipelines and analyzes them for potential bias. Its performance will be recorded and analyzed to determine whether bias can still emerge after data has been cleaned. Through these generated performance data, the system supports both the collection of relevant data and the evaluation of the proposed solution. The following tools will be used to develop and evaluate the system:

1. **Python:** Serves as the primary programming language for developing the tracking decorators, simulating the data pipelines, and executing the Hybrid WOA-DE algorithm. Raschka et al. (2020) point out that Python has become one of the most popular languages for scientific computing, data science, and machine learning, striking a good balance between performance and productivity by building on low-level libraries and clean APIs. This ecosystem is so well adapted to these activities that Python has gained steady adoption in the scientific community over the last decade. With the increasing popularity of the language, numerous modern machine and deep learning tools have been mainly created for the Python ecosystem.

The following libraries will be used for measuring performance:

- a. **time** - The `perf_counter()` function of the time library is used to measure the total execution time of the system. This function is unaffected by system clock changes. This ensures accurate and consistent performance measurement.

- b. **psutil** - This library is used to measure overall system memory consumption. It accesses the operating system's process information to track the total system memory allocated to the entire system during execution.

2. **PostgreSQL(JSONB):** Utilized as the database management system. The semi-structured JSONB format is specifically chosen to store the metadata provenance logs, allowing for schema flexibility and high-speed data retrieval during the algorithmic auditing phase. Aji and Utami (2026) noted that PostgreSQL supports JSON data natively. Their study further revealed that JSONB consistently outperforms the column-based approach in query execution speed across every tested scenario. Additionally, JSONB had a smaller storage size compared to the column-based approach. This will allow the system to deliver results faster while consuming less storage space.
3. **Git:** Employed for version control to maintain the integrity of the system's development lifecycle.
4. **Experiment Sheet:** A paper used to evaluate and record the model's performance and traceability on each data preprocessing pipeline, which will also serve as a guide to answer the research questions.

Data Generation/Gathering Procedure

The research utilized a secondary data gathering procedure by acquiring an existing structured dataset from the study of Ding et al. (2022). The selection was based

on purposive sampling criteria, where the data is required to contain the most common demographic attributes: age, sex, and race.

1. Dataset Formatting

The researchers will first perform a structural simplification of the raw Public Use Microdata Sample (PUMS) variables to ensure the dataset is suitable for auditing. The researchers will rename technical column headers, such as PINCP (Total person's income) and AGE (Total person's age), into human-readable labels to facilitate clearer interpretation in the audit reports. Since the raw rows are recorded in numeric form, the researchers will utilize the PUMS Data Dictionary to map these codes to their corresponding categorical values, for example, converting numerical sex and race codes into their specific string equivalents.

2. Protected Attributes and Binary Target Variable

The researchers identify Age, Sex, and Race as the primary protected attributes for this study. These variables will be monitored to detect demographic or intersectional bias introduced during the cleaning process. To enable the calculation of fairness metrics such as Statistical Parity and Disparate Impact, the researchers will establish a binary target variable based on the income attribute. The researchers define a favorable outcome as an annual income exceeding \$50,000, while any income below this threshold is recorded as a negative outcome.

3. Splitting Dataset into Different Sizes

The entire dataset will be divided into certain portions: 10,000 rows, 100,000 rows and 1,000,000 rows to evaluate the fairness, scalability and reliability of the Hybrid WOA-DE algorithm.

This logarithmic method allows researchers to test whether the tool has consistent performance in the large-scale real-world data environment. This is also supported by Joseph and Vakayil (2021) who showed that the dataset-splitting strategy can impact the worst-case testing performance.

Ethical Considerations

The researchers ensure that the following will be observed throughout the study:

1. Privacy and Confidentiality

All data and information collected and used will be handled responsibly and in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173). Any personal information collected by the system will be confidential.

2. Responsible Use of Data

All data used in the development, testing, and evaluation of the bias auditing tool will be strictly for academic and research purposes only. Sensitive information will only be accessible to authorized persons, including the researchers.

3. Proper Acknowledgment of Sources

All intellectual property, sources of information, related literature and studies, software tools, and programming libraries used in system development will be acknowledged through proper APA-7 citation and referencing to avoid plagiarism.

4. Ethical Use of the System

The system developed in this study is strictly for its stated purpose only. It will not be designed or used in ways that it could cause harm, violate privacy, misuse information, or compromise security.

Data Analysis

Phase 1: Performance Evaluation

This phase focuses on the empirical validation of the hybrid WOA-DE algorithm's efficiency and effectiveness. The goal is to compare the performance of this hybrid system against standalone versions of the Whale Optimization Algorithm and Differential Evolution.

1.1. Fairness

Fairness assessment requires an established ground truth to verify the operational integrity of the auditing instrument. Because real-world bias is inherently silent or latent within raw data, it often remains undetected by standard surface-level evaluations. By introducing controlled bias, researchers can

calculate detection accuracy by comparing the system's generated audit flags against known injected bias.

Procedure: To empirically measure fairness, the researchers will execute bias injection via two distinct methodological approaches:

- Pipeline Transformation Manipulation:** According to Black et al. (2023) highlight data preprocessing, such as handling missing data, data encoding, and feature engineering, as a key stage of consequential design choices where bias can be introduced. Accordingly, the bias is injected in the preprocessing step by the synthetic injection of data inconsistencies and outliers. Researchers may, for example, fill a given demographic category, such as "Male", with inconsistent string values (e.g., "m", "MALE", "M") or extreme numerical outliers to represent a scenario where the automated cleaning steps might mistakenly delete or misinterpret such records. This process enables the study to assess the capacity of the system to detect the representation bias that is produced. Akpinar et al. (2022) explicitly conceptualise a "sandbox tool" to "bias(stress)-test" algorithmic interventions by creating a controlled environment that "counterfactually injects biases in the ML pipeline."
- Binary Target Variable Manipulation:** This involves the intentional changing of binary target variables. For instance, the researchers may change positive results to negative for a specific percentage of a subgroup. This allows the system to verify if the

algorithm can accurately pinpoint the exact moment the fitness score exceeds the default fairness threshold. Chang et al. (2020) show that an attacker with the ability to manipulate the labelling of a subset of training data can drastically increase the test fairness gap of the model, preventing the model from transferring the guaranteed training fairness to test data. Using adversarial labelling, the attacker can selectively flip the labels of data points from a “certain part of the population”, resulting in a “more powerful attack”. The purpose of this “more powerful attack” is to intentionally distort the data distribution and force the fair algorithm to reduce its overall accuracy across the whole distribution to equalise its behaviour.

The objective of the injection is to create a known global optimum within the metadata search space. Injected bias must represent the highest fitness score in the dataset because the metaheuristic search agents are designed to maximize their fitness function. This ensures that successful convergence on the injected bias serves as numerical proof of the algorithm's fairness.

To confirm that the bias injection is successfully realized and to rigorously assess the detection accuracy, convergence reliability, and root-cause diagnostic capabilities of the proposed Hybrid WOA-DE algorithm against standalone WOA and DE, an empirical ground truth benchmark is established across the preprocessing pipeline. Because silent demographic disparities are embedded within high-dimensional intersectional combinations, this reference baseline serves as the definitive evaluation standard for all experimental trials.

1.2. Scalability and Resource Efficiency

This phase evaluates the system's ability to maintain functional performance as the volume of metadata increases. The researchers seek to determine if the Hybrid WOA-DE algorithm can process large-scale datasets without exceeding reasonable time or hardware limits.

Procedure (Execution Time): To measure Audit Latency, the researchers will conduct a series of tests using the ACSIncome dataset. The audit will be performed in logarithmic increments, specifically processing 10K, 100K, and 1M rows of metadata. For each increment, the system will record the total seconds required to complete the search for a bias. This allows the researchers to observe how the execution time grows relative to the data size.

Procedure (Memory Usage): To evaluate the system footprint, the researchers will use the psutil Python library to monitor the peak memory usage during each audit cycle. Monitoring memory is necessary to verify that the hybrid approach remains stable and does not cause a system-crash when navigating high-dimensional JSONB logs.

The objective of these measurements is to provide a realistic assessment of the system's efficiency. By recording these numbers, the researchers can demonstrate whether the metadata-based audit approach is a practical solution for large-scale data pipelines.

The results will be organized into a comparison table showing the relationship between dataset size, time, and memory. The researchers will look

for a linear or stable growth pattern, rather than an exponential one, to prove the system is scalable.

1.3 Reliability

In this phase, the system's ability to consistently and correctly identify bias across different dataset configurations is assessed. The researchers evaluate whether the Hybrid WOA-DE algorithm produces precise and dependable results when exposed to varying levels of injected bias.

Procedure: To measure detection reliability, the researchers will perform multiple audit runs across the different dataset sizes used in the scalability tests, specifically 10K, 100K, and 1M rows. For each run, the system's generated audit flags will be compared against the known injected bias to classify each result as a True Positive, False Positive, True Negative, or False Negative. These classifications will be organized into a Confusion Matrix, which will then serve as the basis for computing the Precision, Recall, and F1-Score of the system.

The objective of this phase is to provide a realistic evaluation of the system's detection consistency. By analyzing these results, the researchers can demonstrate whether the Hybrid WOA-DE algorithm produces dependable results regardless of dataset size or bias type, rather than performing well only under specific conditions.

Phase 2: Root-Cause Traceability Success Rate

This phase addresses the research objective by evaluating the effectiveness of the hybrid algorithm in performing root-cause analysis. The researchers seek to

determine if the system can successfully navigate the metadata provenance to pinpoint the specific stage where the silent bias originated during data preprocessing.

Procedure: The researcher will use the records of the bias injection from Phase 1 as the established ground truth for tracing. For each instance where the Hybrid WOA-DE algorithm detects a bias area, the researcher will extract the corresponding transformation name (e.g., "Handle Outliers" or "Remove Duplicates") from the metadata logs. This identified stage will then be compared against the documented injection log to verify if the algorithm accurately traced the bias to the correct pipeline step. A correct trace will be recorded only if the algorithm precisely pinpoints the exact transformation where the ethical distortion was introduced.

The objective is to calculate the traceability rate to provide a numerical evaluation of the system's diagnostic performance. By comparing the number of correctly identified root causes against the total number of detected biases, the researcher can determine the feasibility of using metadata provenance for ethical auditing in large-scale pipelines.

Table 1: Data Analysis

Research Problem	Data Analysis
Research Problem #1.1: What is the performance of the hybrid WOA-DE algorithm compared to standalone WOA and DE	Statistical Parity Difference (SPD) $SPD = P(\hat{Y} = 1 A = u) - P(\hat{Y} = 1 A = p)$ Disparate Impact

algorithm in terms of fairness of the cleaned dataset?	$DI = \frac{P(\hat{Y} = 1 A = u)}{P(\hat{Y} = 1 A = p)}$ Fitness Function $f(X) = SPD + 1 - DI$									
Research Problem #1.2: What is the performance of the hybrid WOA-DE algorithm compared to standalone WOA and DE algorithm in terms of scalability of increasing data volume?	Execution Time = End Time – Start Time RAM Peak Memory Usage: psutil									
Research Problem #1.3: What is the performance of the hybrid WOA-DE algorithm compared to standalone WOA and DE algorithm in terms of tool reliability?	Confusion Matrix <table><tr><td></td><td>Flagged as Worst</td><td>Ignored</td></tr><tr><td>Worst Bias Stage</td><td>TP</td><td>FP</td></tr><tr><td>Non-Critical Stages</td><td>FN</td><td>TN</td></tr></table> $\text{Precision} = \frac{TP}{TP + FP}$		Flagged as Worst	Ignored	Worst Bias Stage	TP	FP	Non-Critical Stages	FN	TN
	Flagged as Worst	Ignored								
Worst Bias Stage	TP	FP								
Non-Critical Stages	FN	TN								

	$\text{Recall} = \frac{TP}{TP + FN}$ $\text{F1} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$
Research Problem #2: What is the traceability success rate of the metadata provenance system in correctly identifying the root cause of the detected bias?	$\text{Traceability Rate} = \frac{\text{Total Number of Successful Runs}}{\text{Aggregate Runs}}$

Statistical Treatment

Bias detection measurements are highly sensitive to experimental configurations, where small methodological changes can strongly influence the resulting bias scores (Husse & Spitz, 2022). In particular, evaluations with small sample sizes are often unstable. Small subgroups are not representative for concepts in a comprehensive way and are not statistically significant (Husse & Spitz, 2022).

This study enforces a minimum sample size threshold of $N \geq 30$ to reduce the risk of high noise and volatile selection rates that correspond to random variations rather than systemic model behaviour. This threshold is based on the Central Limit Theorem which states that a sample size of at least 30 is the standard guidelines required for a sampling distribution to achieve a normal approximation, assuming the underlying population is not extremely skewed (Diez et al., 2022). The proposed WOA-DE auditor prunes these small subgroups to avoid that the process gets stuck on local-optimum

created by random statistical artefacts, but maximizes only for biases that are structurally and statistically significant in the dataset.

The following methods will be implemented to answer the research questions:

1. Fitness function

A fitness function is a mathematical evaluation tool that measures exactly how "good," "fit," or successful a particular solution or state is. It is the specific mechanism used to guide the algorithm's search through a complex problem. The fitness function will be the combined formula of Statistical Parity Difference and Disparate Impact:

$$f(X) = |StatisticalParityDifference| + |1 - DisparateImpact|$$

If the data is perfectly fair, SPD will be equal to and DI will be equal to 1. The equation becomes 0 otherwise if the data has bias the result will be > 0 .

According to Carey and Wu (2023), a broader definition of demographic parity would involve comparing the ratio of two groups rather than their differences. To conform to the four-fifths requirement of disparate impact law, this ratio formulation was purposefully created. The ratio shows how different things are relative to one another, but the difference formulation shows how big the gap is in terms of actual results. Thus, the need for both mathematical representations depends on the particular context of the fairness evaluation.

Though Statistical Parity and Disparate Impact can be used on any categorical variable in a dataset (Bird et al., 2020), they are primarily defined and applied to protected attributes like race, sex, or age to detect unfair discrimination in AI models or decision systems (La Malfa et al., 2024).

1.1. Statistical Parity

Statistical parity (also known as Demographic parity) is a definition of fairness in machine learning and data analysis where the probability of a positive outcome is equal across all protected demographic groups. According to Osborne and Usani (2026), statistical parity difference measures whether different groups receive predictions at similar rates. A large SPD indicates unequal treatment. It ensures the algorithm's decisions are independent of sensitive attributes, aiming to prevent discrimination by equating selection rates.

To satisfy statistical parity, the probability of a positive prediction must be equal across groups. Mathematically, this is expressed as:

$$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = b)$$

Where:

- $\hat{Y} = 1$ represents the predicted positive/favorable outcome.

- A represents the sensitive demographic attribute (e.g., race or sex).
- a and b represent the distinct groups being compared

When actually auditing a dataset, algorithms compute the Statistical Parity Difference (SPD) to quantify exactly how large the bias gap is:

$$SPD = P(\hat{Y} = 1 | A = u) - P(\hat{Y} = 1 | A = p)$$

Where:

- $A = u$ represents the unprivileged group
- $A = p$ represents the privileged group
- $P(Y=1 | A = u)$ is the probability of a positive prediction for the unprivileged group.
- $P(Y=1 | A = p)$ is the probability of a positive prediction for the privileged group.

In this paper, the privileged group is the intersectional demographic grouping defined by the combination of protected factors, such as age, race, and sex, that receives the highest selection rate of favorable results among a set of data preparation transformation steps. The unprivileged group is made up of all other intersectional demographic groupings in a transformation phase that have a lower selection rate of

favorable outcomes than the privileged group. This operational definition is based on proven algorithmic fairness literature.

First, the analysis of the combination of protected qualities is concerned with 'intersectional bias', how overlapping identities are compounded systematic disadvantages (Castelnovo et al., 2022, Chen et al., 2024). Second, the idea of a privileged or unprivileged group by virtue of being subject to positive or negative treatment is compatible with standard definitions of fairness where the privileged group is positively treated and the unprivileged group is negatively treated (Chen et al., 2023, Hort et al., 2022). Finally, the operational mechanics of the four-fifths rule for disparate impact directly support the dynamic definition of the privileged baseline as the subgroup with the highest selection rate . The four-fifths rule explicitly accounts for disparity by measuring all populations relative to the group with the highest rate of positive outcomes (Castelnovo et al., 2022, Watkins et al., 2022).

The standard formula calculates the difference between the favorable outcome rate of the unprivileged group and the privileged group.

1.2. Disparate Impact

Disparate impact (also known as adverse impact) occurs when a policy, practice, or algorithm that appears completely neutral on the surface disproportionately harms or excludes a protected demographic group. To quantify this bias within a dataset, auditors use the Disparate

Impact metric, which determines the ratio of favorable outcomes that different groups encounter. Osborne and Usani (2026) further define disparate impact as the ratio comparing outcomes between privileged and unprivileged groups.

$$DI = \frac{P(\hat{Y} = 1|A = u)}{P(\hat{Y} = 1|A = p)}$$

Where $P(Y=1| A = u)$ and $P(Y=1| A = p)$ are as defined in the Statistical Parity above.

A ratio of exactly 1.0 means perfect parity, where both groups receive positive outcomes at the exact same rate.

The bias audit flag utilizes a threshold value of 0.2 for ratio-based metrics like disparate impact. According to Castelnovo et al. (2022), the 4/5 rule (or 80% rule) is regularly used as a standard baseline in the literature, stating that the selection rate of any group must be at least 80% of the group with the highest selection rate. The applied validity of this specific threshold is reinforced by Osborne and Usani (2026), who successfully evaluated the fairness of their predictive healthcare framework by ensuring the disparate impact metric remained within the 'acceptable policy range of 0.8 to 1.25'. In the context of our fitness function, this 80% baseline translates to an acceptable deviation tolerance of 0.2, calculated as $|1 - 0.8|$ or $|1 - 1.2|$.

While, for difference-based metrics like statistical parity difference, the threshold is left configurable by the user, with a default value of 0. As Chen et al. (2024) highlight, demanding absolute fairness (such as an SPD of exactly 0) is often too rigid in practice,

therefore, software engineers must empirically establish their own predefined acceptable thresholds to detect the presence of a 'fairness bug' in a specific system.

The final bias audit flag threshold combines the thresholds for disparate impact and statistical parity difference. However, because SPD is configurable, its threshold may be set by the user, with 0 as the default value.

2. Execution Time

This serves as the primary metric for evaluating system scalability. It measures the duration required for the Hybrid WOA-DE algorithm to navigate the JSONB metadata logs and identify the global bias optimum. To ensure precise results, the researchers employ the `time.perf_counter()` function, which provides a steady clock reference unaffected by system time adjustments.

The metric is calculated by determining the difference between audit completion and search initialization. Mathematically, the researchers express this as:

$$\text{Execution Time} = \text{End Time} - \text{Start Time}$$

Where:

- *End Time* is the point at which the algorithm terminates and returns the refined coordinates of the detected bias.
- *Start Time* is the point at which the researcher initializes the algorithm's search agents across the logs.

3. Peak Memory Usage (RAM)

This assessment evaluates the resource efficiency and stability of the auditor during high-volume data processing. By using the psutil library to access real-time process information from the operating system, the researchers can monitor peak RAM usage during execution. The benchmark study of Martins et al. (2025) uses peak resident set size, tracked via psutil, as the standard memory metric for evaluating data processing tools at scales of 1M to 100M records.

This measurement, recorded in Megabytes (MB), helps determine whether the hardware can realistically support auditing datasets that scale logarithmically.

4. Confusion Matrix

A confusion matrix is a square matrix whose size is determined by the number of output classes, where each row captures the predicted class instances and each column captures the actual class instances. It offers a structured summary of a classifier's accurate and inaccurate predictions (Sathyanarayanan & Tantri, 2024). It serves as the foundation for computing the Precision, Recall, and F1-Score of the Hybrid WOA-DE algorithm.

Table 2: Confusion Matrix

	Flagged as Worst	Ignored
Worst Bias Stage	TP	FP

Non-Critical Stages	FN	TN
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Where:

- True Positive (TP) represents the system correctly flagged the bias with the highest score and is in the ground truth.
- False Positive (FP) represents the system flagged a bias that was not in the ground truth.
- True Negative (TN) represents the system correctly did not flag any bias.
- False Negative (FN) represents the system flagged a bias that was in the ground truth but it did not have the highest score.

5. Precision

Precision measures the proportion of correctly flagged positive results out of all instances assigned to the positive class. It is bounded to $[0, 1]$, where 1 indicates that all positive predictions were correct, and 0 indicates that no correct positive predictions were made (Hicks et al., 2022). For this study, precision is used to measure the proportion of bias flags generated by the Hybrid WOA-DE algorithm that were correctly identified as actual bias.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Where TP and FP are as defined in the Confusion Matrix above.

6. Recall

Recall, also known as sensitivity or True Positive Rate (TPR), measures the rate at which actual positive samples are correctly identified, calculated as the ratio of correctly classified positive samples against all samples belonging to the positive class. It is bounded to [0, 1], where 1 represents perfect detection of the positive class, and 0 represents the failure to correctly predict any positive class sample (Hicks et al., 2022). In this paper, recall is used to measure how many of the actual injected biases were successfully detected by the system.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Where TP and FN are as defined in the Confusion Matrix above.

7. F-1 Score

The F1-Score is the harmonic mean of precision and recall. It balances both metrics by penalizing extreme values of either. It is bounded to [0, 1], where 1 represents the highest possible precision and recall values, and 0 represents zero precision and/or recall (Hicks et al., 2022). The F1-Score is used in this study to provide a single summarizing score that balances both precision and recall in evaluating the reliability of the Hybrid WOA-DE algorithm.

$$\text{F-1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where Precision and Recall are the computed values derived from their respective formulas.

8. Traceability Rate

Namli et al. (2024) demonstrate that traceability in AI data pipelines is essential for detecting and attributing bias to specific preprocessing stages. They establish that 'without traceability, it is challenging to detect and mitigate biases.

The percentage of aggregate runs in which the algorithm successfully traces the absolute true global bias hotspot identified.

$$\text{Traceability Rate} = \frac{\text{Total Number of Successful Runs}}{\text{Aggregate Runs}} * 100$$

Where:

- *Total Number of Successful Runs* represents the count of independent runs where the algorithm's global best position successfully converged to and flagged the highest(worst) bias hotspot in the data pipeline.
- *Aggregate Runs* represents the total sample size of execution trials conducted during benchmarking.

9. Friedman Test

To validate the performance of the proposed hybrid WOA-DE against the standalone WOA and standalone DE algorithms, this study employs a two-step non-parametric statistical evaluation framework. First, the Friedman test is utilized to establish an overall performance ranking among all evaluated algorithms across the benchmark functions (Rahimnejad et al., 2023).

Aydilek et al. (2025) define the Friedman test as a "widely used non-parametric method designed to evaluate whether there are statistically significant differences among multiple related groups". This method is highly

appropriate when the same set of algorithms is assessed across different data blocks; it provides a robust solution because it "does not require assumptions of normality or equal variances" and operates strictly on performance rankings.

Accordingly, this study will utilize the Friedman test to mathematically validate whether a globally significant difference exists among the performances of the Hybrid WOA-DE, standalone WOA, and standalone DE algorithms.

$$F_r = \left[\frac{12}{nk(k+1)} \right] \sum_{i=1}^k R_j^2 - 3n(k+1)$$

Where:

- k is the total number of algorithms.
- n is the total number of independent runs.
- R_j is the average rank of the j -th algorithm across all runs.

10. Wilcoxon Signed-Rank test

Because the Wilcoxon signed-rank test is inherently designed for pairwise comparisons, it is subsequently applied in the second step to conduct specific one-to-one evaluations, such as comparing the hybrid WOA directly against the original WOA, and then against the standard DE (Rahimnejad et al., 2023).

This combination of utilizing the Friedman test for overall ranking followed by the Wilcoxon signed-rank test to calculate p-values for statistical significance is widely established as the standard methodology for comparing the

performance of multiple evolutionary and swarm intelligence algorithms (Rahimnejad et al., 2023)

$$W = \sum_{i=1}^{n_r} \text{sgn}[x_{1,i} - x_{2,i}] \cdot R_i$$

Where

- n_r is the number of trials.
- $x_{1,i}, x_{2,i}$ are the observed values for the two algorithms in trial i ,
- R_i is the rank of the absolute difference $|x_{1,i} - x_{2,i}|$
- sgn is the sign of the difference

Due to the inherently stochastic nature of metaheuristic algorithms, the results of a single execution are insufficient for comparative analysis. Following standard methodological guidelines for metaheuristic benchmarking, each algorithm was independently executed 30 times to ensure robust statistical comparisons (Kyrou et al., 2026, Lee & Zhuo, 2021).

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Appendices

Appendix A: Experiment Sheet

- **Dataset Name:** _____
- **Protected Attribute/s:** _____
- **Target Variable:** _____

1. Model Evaluation

Dataset Size	Algorithm	Fairness			Scalability		Reliability		
		SPD	DI	Fitness Function	Execution Time	Peak Memory Usage	Precision	Recall	F-1 Score
10k	WOA-DE								
	WOA								
	DE								
100k	WOA-DE								
	WOA								
	DE								
1m	WOA-DE								
	WOA								
	DE								

2. Traceability

Dataset Size	Preprocessing Stage	No. of Traced Bias	No. of Detected Bias	Traceability Rate
10k	Pipe 1: Handling Missing Values			
	Pipe 2: Handling Duplicates			
	Pipe 3: Handling Outliers			
100k	Pipe 1: Handling Missing Values			
	Pipe 2: Handling Duplicates			
	Pipe 3: Handling Outliers			

1m	Pipe 1: Handling Missing Values			
	Pipe 2: Handling Duplicates			
	Pipe 3: Handling Outliers			